Sentinel AI: product requirements

On-device voice incident reporting · v0.1 (hackathon build 7c6bb7c) · 11 October 2026

1. Problem

When something goes wrong at a workplace (a leak, a stuck lift, smoke, a broken badge reader) the person who notices it rarely knows which team owns it, what details that team needs, or how urgent it is. Reports arrive late, incomplete ("there's a leak somewhere upstairs") and to the wrong desk, and responders spend the first minutes calling back for the location. Typing a form on a phone while standing in water is the wrong interface, and sending a voice recording of an internal site to a cloud service is often not allowed.

2. Product

A voice assistant on the phone that takes an incident report the way a good dispatcher would: the employee says what happened; the assistant works out what kind of incident it is and who handles it, asks only for what's missing (starting with where), and files a structured, prioritised report the moment the essentials are known. It runs entirely on the device (Snapdragon 8 Elite NPU and CPU), so it works offline and no audio or text leaves the phone.

3. Users

User	Needs
Reporter (any employee)	Report in under a minute, hands-free, without knowing department names or form fields; know that the right team has it and how soon they'll respond.
Facilities desk / responder	Complete reports with location and safety facts up front, sorted by urgency, with the right people already notified; mark progress.
Site / EHS admin	Control the incident types, fields, routing rules and SLAs without code (the taxonomy file).

4. Goals and non-goals

Goals

- Natural back-and-forth by voice, reply within about 2 s of the user finishing.
- Correct incident type and routing for clear reports; ask when unsure.
- Never file without WHAT and WHERE; never invent a value.
- Fully on-device and offline.
- Works in a noisy workplace.

Non-goals (v0.1)

- Sending reports to a server or ticketing system.
- Authentication (the role choice is not access control).
- Languages other than English (Hindi planned).
- Photos, maps, indoor positioning.

5. User stories

1. As a reporter, I tap once and describe the problem in my own words, and the assistant asks me follow-up questions out loud until it has what it needs.
2. As a reporter, if I'm vague ("something's wrong with the water"), the assistant offers me the likely options and I pick one by voice.
3. As a reporter, I'm told as soon as my report is ready and which team will handle it, and I can say "done" at any point after that.
4. As a reporter, I can see the form filling itself from what I said and correct a field by tapping it.
5. As a reporter, I can see my past reports and their status.

6. As a responder, I see open reports most urgent first, with location, safety facts, people notified and a respond-by time.
7. As a responder, I can have a report read out to me and mark it in progress or resolved.
8. As an admin, I change incident types, questions, rules and SLAs in one JSON file.

6. Functional requirements

ID	Requirement	Status
F1	Voice-only reporting: one mic control; the mic re-opens automatically after each spoken question; tapping during speech interrupts.	Done
F2	End-of-turn detection on a natural pause (~0.9 s), robust to steady and changing background noise; ignores music, noise and distant talkers.	Done
F3	On-device speech recognition with live caption while speaking.	Done
F4	Classify the incident into one of 28 taxonomy types (or not-an-incident) every turn, with calibrated confidence.	Done
F5	Essentials first: ask only for missing type/building/floor; refuse to file without them, saying why.	Done
F6	Clarify uncertain types with a top-3 menu or an open question; accept "the second one" or a label.	Done
F7	After READY, ask remaining details one question at a time, safety first, ≤ 2 fields per question, each once; "done" files.	Done
F8	Extract field values only when grounded in the user's words; enums from the allowed list; "unknown" only when said.	Done
F9	Route from the taxonomy: owning team, priority (rules can raise it), roles notified, secondary incidents, SLA; respond-by from the device clock.	Done
F10	Spoken replies in a natural voice that starts within ~0.3 s of the text being ready.	Done
F11	Live form view with provenance ("Auto-filled from voice input"), per-field status, tap-to-correct with type validation.	Done
F12	Review screen and filed confirmation with reference ID; history of the user's reports.	Done
F13	Facilities desk: list by status and priority, filters by status and team, detail, read-aloud, start work / resolve.	Done
F14	Out-of-scope requests are declined without filing.	Done
F15	Hindi / Hinglish input with replies in the user's language.	Planned
F16	Sync filed reports to a server / ticketing system when online.	Planned

7. Non-functional requirements

Area	Requirement	Measured (v0.1)
Latency	Reply audible ≤ 2.0 s median and ≤ 3.0 s p90 after the user stops talking	1.78 s median, 2.79 s p90
Processing	≤ 1.2 s median from turn detected to reply audible	1.0 s
Classification	≥ 85 % incident type accuracy on held-out conversations, on the NPU	87.6 %
Transcription	≥ 80 % of clean turns word-perfect; building and floor captured	19/23 (83 %); one building code misheard
Privacy	No network access; no transcripts in system logs	Met
Resources	Fits a 12 GB phone alongside other apps; no thermal throttling over a session	~2.5 GB resident; headroom 0.33–0.40
Startup	Models ready ≤ 3 s after launch	~1.4 s
Accessibility	48 dp targets, font scaling, reduced motion, labeled controls	Met

8. Success metrics

- Time from first word to filed report (target median < 60 s).
- Share of reports filed with building and floor (target 100 %, enforced) and with all safety fields answered.
- Share of reports routed to the right team without reassignment (target ≥ 90 %).
- Responder call-backs per report for missing information (target: halve the current rate).
- Turns per report (target ≤ 4) and abandoned conversations.

9. Experience principles

- **Voice first.** The screen confirms; the conversation leads. No typing to report.
- **Ask less.** Only missing essentials first; never ask the same thing twice; the user can stop at any time once READY.
- **Say what happens next.** "Your report is ready and Facilities Engineering will handle it."
- **Calm authority.** Editorial type, one navy accent, crimson only for urgency and missing essentials.

10. Dependencies and constraints

- Snapdragon 8 Elite class device with Hexagon NPU; QAIRT/SNPE runtime (not redistributable, supplied at build time).
- About 2.2 GB of model files copied to the device; English only in v0.1.
- The incident taxonomy is the single source of truth for types, fields, questions, routing and SLAs.

11. Risks

Risk	Mitigation
Misheard locations (single-letter building names)	Whisper primed with building codes; normalization; essentials re-asked; tap-to-correct on the form.
Bystander speech taken as a turn in busy spaces	Peak-level gate, non-speech filter; option to ignore out-of-scope voice turns instead of closing.
Small LLM misses values	Deterministic rules first, validation, catch-up extraction; fine-tuned Qwen; larger model on the NPU later.
Memory on mid-range phones	Smaller Whisper graph only, quantized models, load on demand.
Another build overwrites the app on shared kits	Separate application id for team builds.

12. Roadmap

1. **Next:** fine-tuned Qwen for extraction; shorter pause rule option (0.6 s); speak fixed lead-ins while the question is generated.
2. **Then:** Hindi/Hinglish via Whisper small (encoder on the NPU) with Hindi question templates and a Hindi voice.
3. **Later:** 3–4B language model on the NPU (Genie); server sync; per-site taxonomies; photos.